Modeling Anthropogenic CO2 Accumulation: A Bayesian Comparison of Linear and Accelerating Trends

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The Mauna Loa Observatory is a premier baseline site for monitoring historical and current atmospheric carbon dioxide (CO2) concentrations. This report investigates whether the long-term increase in atmospheric CO2 is best described by a constant linear increase or an accelerating quadratic trend. Using monthly average CO2 data from the NOAA Global Monitoring Laboratory, we address missing values and compute annual averages to isolate the long-term trend from seasonal variations. We perform a Bayesian analysis to estimate the parameters of both a linear model and a quadratic model, with a specific focus on the acceleration parameter. The two models are then evaluated and compared using an approximate Bayes factor to determine which model better captures the underlying trend dynamics. Finally, utilizing the best-fitting model, we project the annual atmospheric CO2 concentrations for the next decade, providing predictive estimates complete with Bayesian credible bands to quantify uncertainty.

Introduction

See code 1 in Appendix for the implementation.

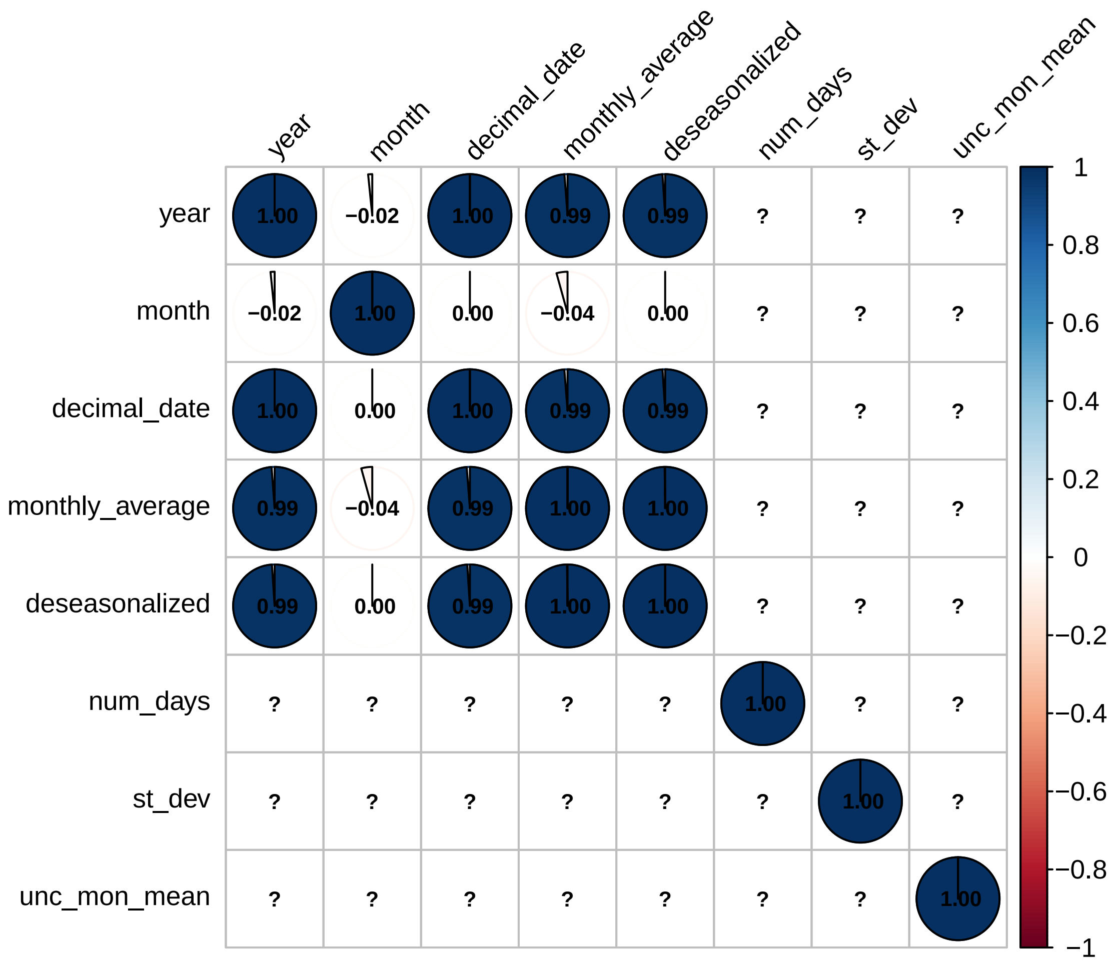

Figure 1: Correlation plot of the variables in the dataset

Scientific Context

Objectives

Data Description and Preprocessing

Dataset Overview

Data Cleaning

Aggregation

Exploratory Visualization

Bayesian Methodology and Models

Model 1: Linear Trend

Model 2: Quadratic Trend

Inference Strategy

Results and Inference

Parameter Estimation

Model Comparison

Predictive Forecasting

Ten-Year Projections

Uncertainty Quantification

Conclusion

Summary of Findings

Limitations and Future Work

References

Appendix: Source code

```r

# This is an example of a test file for EDA (Exploratory Data Analysis)
source("code/Data.R")
library(corrplot)

corrplot(cor(data), method = "pie", addCoef.col = "black", tl.cex = 0.8, number.cex = 0.7,
         tl.col = "black", tl.srt = 45, addrect = 2, rect.col = "blue", rect.lwd = 1.5)
```

Listing 1: Plotting the data